Retail Profitability Analysis: Where Discounts Destroy Profit

A retailer can sell a lot and still lose money. I wanted to find out where the Global Superstore dataset (51,290 order lines, 2011 to 2014) loses money and why. I cleaned the data in Python, analysed it with SQL in Google BigQuery, and built a Power BI dashboard to show the result.

Short answer: no order sold at full price lost money. Once a discount goes above 20%, almost every order does.

The question

Which products and orders drain profit, what is behind it, and what should the business change?

Dataset

Global Superstore (Kaggle). 51,290 order lines, 24 columns, orders from 1 Jan 2011 to 31 Dec 2014, across 7 markets. It includes sales, quantity, discount, profit and shipping cost.

Tools

- Python (Pandas) in Google Colab for cleaning
- Google BigQuery (SQL) for the analysis

- Power BI Desktop for the dashboard
- GitHub

What I did

1. **Cleaned the data.** Checked for nulls and duplicates (none found), dropped the postal code column because 80.5% of it was empty, trimmed text, converted dates, and checked for impossible values. Added profit margin, shipping days, order year, order month and season columns. The cleaning code is in `notebook/01_cleaning.py`.
2. **Loaded it into BigQuery** and built a view, `v_dashboard`, that groups discounts into four bands (none, 1-20%, 21-40%, above 40%).
3. **Ran the analysis in SQL.** Profit by sub-category, margin by discount band for the whole store and for Tables, and the share of order lines that lose money. The queries are in `sql/retail_analysis_final.sql`.
4. **Built the dashboard in Power BI** with KPI cards, charts and filters for market, category, season and year.

What I found

Finding	Number
Total sales / profit	12.64M / 1.47M (margin 11.6%)

Finding	Number
Order lines that lose money	12,544 of 51,290 (24.5%)
Full-price lines that lose money	0
Margin with no discount	25.3%
Margin at 1-20% discount	13.7%
Margin at 21-40% discount	-17.3%
Margin above 40% discount	-74.1%
Lines discounted above 20%	11,328 (22% of lines, 15% of sales)
Loss from those lines	about 815K, equal to 55% of total profit
Share of those lines that lose money	about 90% (98.4% above 40%)
Share of all losing lines that sit above a 20% discount	81%

Tables is the only sub-category that loses money overall: sales of 757K, profit of -64K, an average discount of 29.1%. It is profitable up to a 20% discount (+68K), then loses 133K on the 467 lines discounted above that.

Recommendations

1. Cap discounts at 20% and require manager approval for anything higher.
2. Stop deals above 40% first. Nearly all of them (98.4%) lose money.
3. Review Tables pricing and its discount policy, since it loses money even at 21-40%.

Limitations

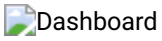
- The dataset has no inventory data, so I could not measure inventory turnover or slow-moving stock.
- The data shows that discounts and losses go together. It cannot prove that the discount alone caused every loss, because costly products may also get bigger discounts.
- The 815K figure is an upper bound on what a discount cap could recover, not a forecast, because some of those sales would not happen at a lower discount.
- I did not analyse seasonality in depth. The dashboard has a season filter and a monthly trend chart for anyone who wants to look further.

Repository structure

```
retail-profitability/
├── data/                cleaned CSV (or a
link to the Kaggle source)
├── notebook/01_cleaning.py
├── sql/retail_analysis_final.sql
├── dashboard/           Power BI file (.pbix)
├── images/              dashboard and query
```

```
screenshots
├── report/                2-page PDF report
└── README.md
```

Dashboard

 Dashboard

How to reproduce

1. Download Global Superstore from Kaggle: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset
2. Run `notebook/01_cleaning.py` in Google Colab to create `global_superstore_clean.csv`.
3. Upload the CSV to BigQuery as `retail.orders_clean`.
4. Run the queries in `sql/retail_analysis_final.sql`.
5. Connect Power BI to the `retail.v_dashboard` view in Import mode.

Mary Rebecca Reena | BCA (Data Analytics), St. Joseph's University, Bengaluru